

1. Administrative & Study Identification

Full Study Title: Phase III, Open-label, Study of First-line Dato-DXd in Combination With Rilvegostomig for Advanced Non-squamous NSCLC With High PD-L1 Expression (TC \$\\ge 50\\%\$) and Without Actionable Genomic Alterations **Short Title/Acronym:** TROPION-Lung10 **Protocol Number:** D7632C00001 **NCT Number:** NCT06357533 **Posting ID:** 304483877631871790 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Datopotamab deruxtecan (Dato-DXd) + Rilvegostomig (AZD2936) **Comparator:** Pembrolizumab (Keytruda, ATC Code: L01FF02) **Phase of Development:** Phase III **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the progression-free survival (PFS) and overall survival (OS) of first-line Dato-DXd plus rilvegostomig versus standard-of-care pembrolizumab in patients with advanced/metastatic nonsquamous NSCLC with high PD-L1 expression. **Secondary Objectives:** To assess the objective response rate (ORR), duration of response (DOR), PFS in the Full Analysis Set (FAS) population, overall survival (OS) in the FAS population, second PFS (PFS2), pharmacokinetics (PK), patient-reported outcomes, and safety profile.

2.2 Background & Rationale While immunotherapy targeting the PD-1/PD-L1 pathway has improved outcomes in advanced NSCLC patients without actionable genomic alterations (AGAs), resistance to treatment remains a critical challenge. Datopotamab deruxtecan (Dato-DXd), a TROP2-directed antibody-drug conjugate, and rilvegostomig, a bispecific anti-PD-1/anti-TIGIT antibody, have shown promising efficacy and manageable safety in NSCLC. This trial investigates if this combination can overcome resistance mechanisms and improve survival outcomes compared to pembrolizumab monotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 675 evaluable patients **Number of Sites:** Global Multicenter **Estimated Study Start:** 2024 **Estimated Primary Completion:** To Be Determined

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults with nonsquamous stage IIIB/C or IV metastatic non-small cell lung cancer (NSCLC) not amenable to curative surgery or definitive chemoradiation. 2. Treatment-naïve for locally advanced or metastatic disease. 3. High PD-L1 expression (Tumor cell [TC] $\geq 50\%$) and no actionable genomic alterations (AGAs).

4.2 Exclusion Criteria 1. Clinically significant third-space fluid retention (e.g., pleural effusion) that is not amenable for repeated drainage. 2. History of any interstitial lung disease (ILD)/pneumonitis, including non-infectious drug-induced or radiation pneumonitis requiring steroids, or suspected ILD/pneumonitis at screening. 3. Any evidence of severe or uncontrolled disease that makes it undesirable for the participant to participate or jeopardizes protocol compliance.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Intravenous administration in 21-day cycles. Patients are randomized to receive either Dato-DXd in combination with rilvegostomig, rilvegostomig monotherapy, or pembrolizumab (Trade Name: Keytruda, ATC Code: L01FF02) monotherapy. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology patients as per institutional guidelines. **Prohibited:** Concurrent administration of other investigational agents, anticancer therapies, or immunosuppressive treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Tumor biopsy/Biomarker testing (PD-L1/TROP2), Baseline Scans
Baseline	Day 0	Randomization, First Dose	Treatment
Every 3 Weeks	Investigational Product administration, Safety Labs, Efficacy Assessment	End of Study	Variable
Final Vital Signs, Exit Interview, Off-treatment safety follow-up			

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) and Overall Survival (OS) in TROP2 biomarker-positive participants. **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) in the FAS population. * Overall Survival (OS) in the FAS population. * Assessment of Objective Response Rate (ORR) by BICR in TROP2 biomarker positive and FAS populations. * Second Progression-Free Survival (PFS2) and Pharmacokinetics (PK). * Patient-reported lung cancer symptoms and physical functioning.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each treatment cycle, specifically focusing on specific AEs of interest such as Interstitial Lung Disease (ILD)/pneumonitis. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for severe or life-threatening events. **Data Monitoring Committee (DMC):** Independent external committee responsible for ongoing safety evaluations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Full Analysis Set (FAS) for efficacy endpoints; Safety Analysis Set for adverse events and tolerability. **Sample Size Justification:** Target enrollment of ~675 patients powered to detect a statistically significant difference in dual primary endpoints (PFS and OS). **Handling of Missing Data:** Methods such as censoring at the last evaluable tumor assessment for PFS analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Secure electronic data capture (EDC) systems with full audit trails for eCRF locking and query resolution.

11. Study Identifiers & Governance

Primary ID: D7632C00001 **Registry Number:** NCT06357533 **Posting ID:** 304483877631871790 **Sponsor/Lead Organization:** AstraZeneca **Study**

Title: TROPION-Lung10 **Official Regulatory Title:** A Phase III, Randomised, Open-label, Global Study of Datopotamab Deruxtecan (Dato-DXd) in Combination With Rilvegostomig or Rilvegostomig Monotherapy Versus Pembrolizumab Monotherapy for the First-line Treatment of Participants With Locally-advanced or Metastatic Non-squamous NSCLC With High PD-L1 Expression (TC $\geq 50\%$) and Without Actionable Genomic Alterations (TROPION-Lung10)

12. Clinical Framework (NSCLC Specific)

Primary Condition: Metastatic Non-Small Cell Lung Cancer (NSCLC)
Disease Area: Non-Small Cell Lung Cancer **Preferred Disease Name:** Metastatic non-small cell lung cancer **Preferred UMLS Name:** PD-L1 Positive Non-Small Cell Lung Carcinoma **Diagnosis Threshold:** Stage IIIB/C or IV with high PD-L1 expression (TC $\geq 50\%$) without Actionable Genomic Alterations **Medical Methodology:** First-Line Immunotherapy and TROP2-directed Antibody-Drug Conjugate
Optimization Target: Progression-Free Survival and Overall Survival Extension

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced NSCLC infused therapy **Education Protocol (HCP):** Site initiation visits and continuous protocol training regarding ILD/pneumonitis recognition and management **Education Protocol (Patient):** Comprehensive informed consent and AE self-reporting education **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for tumor scans to ensure unbiased efficacy measurements

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until death, withdrawal, or study closure **Onsite Visit Cadence:** Every 3 weeks (Q3W) aligned with treatment cycles **Remote Monitoring Frequency:** Inter-cycle continuous monitoring for safety endpoints **Checklist Review Interval:** Regular interval scans (e.g., every 6-9 weeks) for RECIST v1.1 evaluation

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				